

CALAFAT, Romania

WMO: 154820

Lat: 43.9850N

Lon: 22.9461E

Elev: 62

StdP: 100.58

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-8.7	-13.4	1.2	-10.8	-10.3	1.6	-7.0	15.5	2.0	9.7	3.8	2.0	260	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	35.1	21.5	33.6	21.2	32.0	21.3	24.5	30.3	23.4	29.6	22.7	29.0	2.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	17.6	27.3	21.6	16.4	26.2	20.7	15.5	25.3	74.1	30.7	70.1	29.3	67.1	29.1	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.1	7.0	DB	-16.2	37.6	4.5	2.2	-19.4	39.2	-22.0	40.5	-24.5	41.7	-27.8	43.3	(4)
(5)			WB	-16.3	25.3	4.4	2.1	-19.5	26.8	-22.1	28.0	-24.6	29.1	-27.8	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.1	4.2	8.2	13.3	18.3	22.7	24.5	24.4	19.6	12.7	6.3	1.9		
	DBStd	9.40	5.00	5.26	4.21	3.92	3.29	2.92	2.55	2.92	3.93	4.13	4.27	4.76		
	HDD10.0	979	313	170	82	15	0	0	0	0	0	21	126	253		
	HDD18.3	2566	571	395	314	156	43	3	1	2	34	177	362	509		
	CDD10.0	2089	1	8	27	114	257	380	449	445	288	106	13	3		
	CDD18.3	635	0	0	0	4	41	134	192	189	71	4	0	0		
	CDH23.3	6285	0	0	6	71	375	1274	1938	1931	639	51	0	0		
	CDH26.7	2714	0	0	0	14	83	517	891	955	247	8	0	0		
(14) Wind	WSAvg	2.4	2.1	2.8	3.0	3.1	2.5	2.4	2.4	2.1	2.2	2.2	2.1	2.5		
(15) Precipitation	PrecAvg	562	39	36	39	48	56	54	53	42	49	50	44	51		
	PrecMax	986	77	85	88	129	115	130	124	171	143	150	103	132		
	PrecMin	276	7	1	4	0	23	9	5	1	3	4	0	0		
	PrecStd	148	21	26	25	25	25	31	35	39	36	37	29	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	19.6	24.1	28.8	30.7	35.7	36.4	37.7	34.1	27.5	21.3	15.6		
		MCWB	9.0	11.9	13.3	17.9	21.0	23.6	22.1	21.9	19.7	18.7	15.1	9.9		
	2%	DB	10.8	16.5	20.3	25.3	28.9	33.2	34.7	35.4	31.7	24.5	17.3	12.6		
		MCWB	7.1	10.5	12.0	17.2	19.6	22.0	21.6	21.1	19.9	16.8	12.7	8.4		
	5%	DB	8.0	14.3	17.7	22.6	27.2	31.4	33.2	33.7	29.5	22.1	14.4	9.8		
		MCWB	5.4	9.8	10.7	15.2	18.7	21.8	21.5	20.8	19.2	15.5	11.4	6.6		
	10%	DB	6.0	11.9	15.3	20.5	25.4	29.7	31.5	32.0	27.3	19.9	12.0	8.0		
		MCWB	4.3	8.4	9.4	14.1	18.0	21.2	21.6	20.9	18.9	15.1	10.5	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	13.5	14.4	19.1	23.8	25.1	25.9	25.9	24.3	21.4	15.1	10.7		
		MCDB	13.7	16.9	20.6	25.7	29.4	31.8	30.6	32.3	30.1	23.2	19.4	13.8		
	2%	WB	7.4	11.6	12.7	17.8	21.3	23.7	24.6	23.7	22.2	17.6	13.4	8.8		
		MCDB	10.2	15.4	19.4	24.3	26.3	30.2	29.5	30.0	28.0	22.1	16.2	12.4		
	5%	WB	5.9	10.0	11.3	16.3	20.0	23.0	23.5	22.8	20.6	16.2	12.1	7.2		
		MCDB	7.3	13.7	16.9	21.7	25.3	29.1	29.5	29.6	26.2	21.3	13.9	9.4		
	10%	WB	4.6	8.5	10.1	15.0	18.6	22.2	22.5	21.8	19.3	15.1	10.6	5.9		
		MCDB	6.0	11.5	14.6	19.4	23.7	27.9	29.0	28.7	24.9	19.4	11.9	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.9	10.0	10.7	10.9	11.9	12.2	13.3	11.7	10.0	6.0	6.3		
		MCDBR	10.3	12.1	15.0	15.2	14.1	14.8	15.5	16.5	15.5	14.7	10.8	9.7		
	5% WB	MCWBR	7.2	7.7	8.2	7.7	6.2	5.4	4.8	5.1	5.5	7.4	6.9	6.4		
		MCWBR	9.1	11.2	13.6	13.8	12.7	13.3	12.4	13.4	13.3	13.5	8.7	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.374	0.413	0.474	0.454	0.448	0.459	0.477	0.446	0.403	0.372	0.346		
	taud		2.288	2.204	2.168	2.019	2.143	2.200	2.170	2.117	2.168	2.278	2.315	2.312		
	Ebn at Noon		755	796	814	792	823	828	812	781	772	756	718	715		
	Edh at Noon		86	111	131	164	149	142	144	147	130	102	84	77		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.23	3.39	4.58	5.59	6.29	6.35	5.65	4.05	2.64	1.53	1.18		
	RadStd		0.21	0.32	0.43	0.52	0.49	0.52	0.42	0.49	0.35	0.37	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.59	N/A	N/A	N/A	+0.40	N/A	N/A	-149	+93	N/A

Nomenclature: See separate page